Estd.1917

Department of CSE, U.C.E, O.U
CLASS TEST – I (MARCH, 2026)
B.E (CSE and AI&ML) VI Semester
(Open elective offered by Mechanical Engg. Dept.)
Subject: 3D PRINTING TECHNOLOGY

Estd.1929

Test Duration: 1.5 hours **Maximum Marks: 30**

Note: i. First Question is compulsory and answer any two questions from the remaining.
ii. Missing data, if any, may suitably be assumed

Q. No		Question	Marks	CO No.	BT Level
1	a	**Distinguish** between 3D Printing and Conventional Manufacturing process	2	I	II
	b	**Compare** the parallels between geometric modelling and Rapid Prototyping	2	I	II
	c	**Pair** the following 3D Printing Technologies with their correct working principle (A) SLA (i) Material extrusion process (B) LOM (ii) Material jetting process (C) Polyjet (iii) Photo polymerisation (D) FDM (iv) Sheet Lamination (v) Powder based AM technology	2	II	I
	d	**Differentiate** between Polyjet and FDM	2	II	II
	e	**Classify** various types of Powder bed fusion 3D Printing Technologies	2	III	II
2	a	**Illustrate** with a neat sketch the classification of various 3D Printing Technologies	5	I	II
	b	"Even though there is an increase in product complexity 3D Printing helped in reducing Project Completion time". **Justify** the above statement with a neat sketch.	5	I	V
3	a	**Discuss** the working principle of FDM with a neat sketch. Mention its advantages, limitations and applications.	7	II	II
	b	**Applying** which most appropriate 3D Printing Technology we can print the following: (i) Jewellery (ii) Multi coloured parts and (iii) Cell phone casing	3	II	III
4	a	Briefly **explain** with a neat sketch the 3D Printing process chain.	6	I	II
	b	With a neat sketch, **Outline** the working of principle of SLS.	4	III	II

Note: CO: Course Outcome, BT: Bloom's Taxonomy